

CIS 4374 Project Management

Smart Parking Platform

Homework 2 Work Breakdown and Scheduling

Initial software landscape, vision and scope, and software requirements specification

Reza Kazmi

Organization: WE ARE \$oftware corp

Date: September 17, 2026

1. Research on Existing Parking Applications

ParkMobile allows drivers to pay for parking and make reservations. It also offers management tools for parking providers. A strength is its support for both drivers and operators. However, its website does not specify a consistent live availability update across facilities.

SpotHero helps drivers compare parking locations and prices, reserve parking, and access a parking pass. Its straightforward booking process is a strength. One limitation is that modification and cancellation options depend on the reservation terms.

ParkWhiz allows drivers to compare prices, reserve parking, and prepay before arrival. Its mobile parking pass simplifies entry. However, the reviewed page does not explain how frequently occupancy information updates or how delayed information is displayed.

Our proposed difference: The Smart Parking Platform will emphasize clearly labeled availability updates, reservation management, and operator reporting. Drivers will see when availability was last updated and receive a warning when information is outdated. These are proposed priorities, rather than claims that competitors lack these features.

2. Vision and Scope

Project Background and Company Intro

For this fictional project, we assume a group of parking garage operators approached WE ARE \$oftware \$orp to plan a system that improves parking search and reservation management. WE ARE \$oftware \$orp is a fictional software development studio with unlimited theoretical resources and budget. The project must still follow the semester timeline.

Business Need

Drivers can spend unnecessary time searching for parking, causing frustration and congestion. Parking operators also need tools to monitor occupancy and manage available spaces.

The Smart Parking Platform will help drivers locate, reserve, and pay for parking through web and mobile applications. It will provide directions, reservation notifications, and receipts. Operators will use a dashboard to manage availability, prices, and occupancy reports.

Initial Scope

The proposed system includes:

- User accounts and vehicle information

- Parking search, interactive maps, and current availability
- Reservations, digital payments, history, and receipts
- Notifications and navigation assistance
- Operator tools for availability, pricing, and reporting

The initial scope excludes garage construction, sensor manufacturing, physical maintenance, and parking enforcement.

3. Initial SRS and Use Cases

Purpose and Users

This draft identifies the system's initial requirements. Drivers will use the web and mobile applications, while parking operators will use a management dashboard. Requirements will be expanded as stakeholder needs to become clearer.

Functional Requirements

The system shall allow drivers to manage accounts, find parking, make reservations, pay, and review their parking activity. It shall also allow authorized operators to manage availability and prices and view occupancy reports.

Proposed Nonfunctional Requirements

These initial targets will be reviewed later:

- Parking search results should be loaded within two seconds under an agreed test workload.
- Availability of readings older than 60 seconds should be marked as outdated.
- Users should only access records and functions permitted for their role.

Initial Use Cases

- **Register an Account**
A new driver enters account information. The system validates the details and creates an account after verification.
- **Log In**
A registered user submits valid credentials. The system opens the appropriate driver or operator interface.
- **Manage Profile and Vehicles**
A logged-in driver updates contact details or vehicle information. The system validates and saves the changes.

- **Search for Parking**
A driver enters a destination and parking period. The system displays matching garages.
- **View Parking on a Map**
A driver opens the map and selects a garage marker. The system displays that garage's information.
- **View Current Availability**
A driver selects a garage. The system displays reported availability and its latest update time.
- **Reserve Parking**
A logged-in driver selects a garage, time, and vehicle. The system checks inventory and confirms the reservation after successful payment.
- **Modify a Reservation**
A driver requests changes to an existing booking. The system checks availability and policy restrictions before saving approved changes.
- **Cancel a Reservation**
A driver cancels an eligible booking. The system releases the reserved capacity and records any applicable refund.
- **Pay for Parking**
A driver submits payment during checkout. The system records the payment provider's approval or failure of results.
- **View History and Receipts**
A logged-in driver selects a past reservation. The system displays its details and provides a receipt for completed payments.
- **Receive Notifications**
A reservation event occurs. The system sends a confirmation, reminder, or update through the driver's approved notification channel.
- **Navigate to a Garage**
A driver selects Directions. The system opens a mapping service with the garage entrance as the destination.

- **Manage Availability and Prices**
An authorized operator updates garage availability or rates. The system checks for conflicts and saves valid changes.
- **View Occupancy Reports**
An authorized operator selects a garage and date range. The system displays occupancy information and identifies missing data.

4. Work Breakdown Structure

The Work Breakdown Structure divides the Smart Parking Platform into three levels: the overall platform, its major components, and the features or deliverables within each component.

1.0 Smart Parking Platform

1.1 Project Planning

- 1.1.1 Vision and Scope document
- 1.1.2 Software Requirements Specification
- 1.1.3 Project schedule and stakeholder updates

1.2 System Design

- 1.2.1 Web and mobile interface designs
- 1.2.2 Database and API design
- 1.2.3 Payment, mapping, and garage integration designs

1.3 User Accounts

- 1.3.1 Registration, login, and password recovery
- 1.3.2 Driver profiles and vehicle records
- 1.3.3 Operator accounts and access permissions

1.4 Parking Search and Garage Management

- 1.4.1 Parking search and interactive map
- 1.4.2 Garage details, availability, and navigation
- 1.4.3 Operator controls for garage information, availability, and pricing

1.5 Reservations and Payments

- 1.5.1 Reservation creation, modification, and cancellation
- 1.5.2 Payment processing and refunds

1.5.3 Reservation history and receipts

1.6 Notifications and Reporting

1.6.1 Reservation confirmations and reminders

1.6.2 Occupancy reports and graphs

1.6.3 Occupancy report exports

1.7 Testing and Release

1.7.1 Functional and integration test results

1.7.2 Security, usability, and user acceptance test results

1.7.3 Release package and user instructions

5. Draft Timeline and Gantt Chart

This draft assumes a 12 week development period from September 21 through December 13, 2026. The dates and durations are preliminary estimates for the fictional development project. They are not the course submission schedule and will be revised as requirements become clearer.

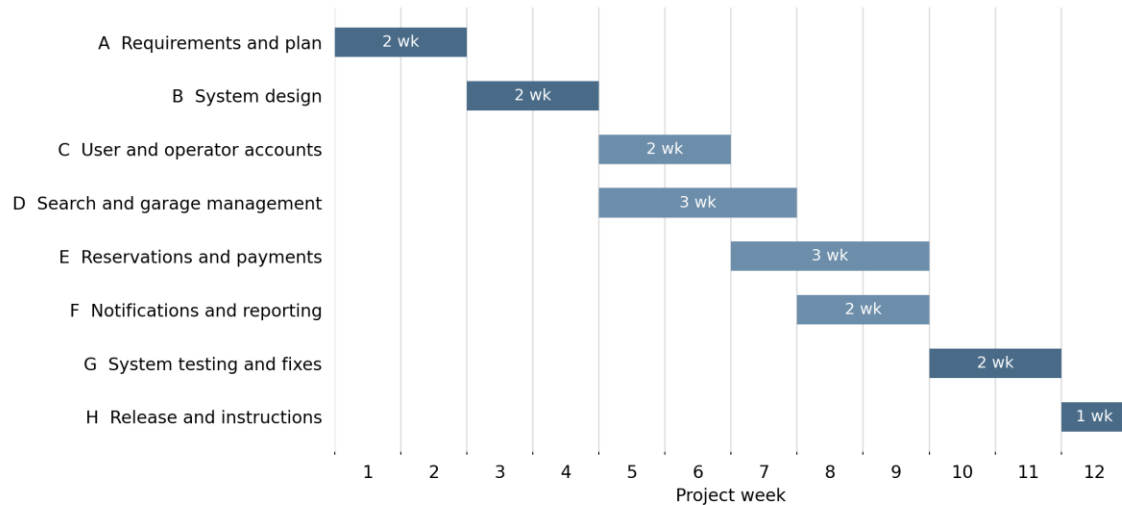
Separate development teams are assumed to work on some components at the same time. Each development activity includes component testing. Project coordination, document revisions, and stakeholder updates continue throughout the schedule.

ID	Activity	WBS	Weeks long	Planned weeks	Depends on
A	Requirements and initial plan	1.1	2	1-2	None
B	Interfaces, database, and integration design	1.2	2	3-4	A
C	User and operator accounts	1.3	2	5-6	B
D	Parking search and garage managements	1.4	3	5-7	B
E	Reservation and payments	1.5	3	7-9	C
F	Notifications and reporting	1.6	2	8-9	C, D
G	System testing and defect resolution	1.7.1-1.7.2	2	10-11	E, F
H	Release and user interactions	1.7.3	1	12	G

Gantt Chart

Smart Parking Platform Draft Schedule

Proposed development period: September 21–December 13, 2026



Major Milestones

- End of Week 2: Initial requirements and project plan ready for review.
- End of Week 4: System design complete.
- End of Week 9: Planned features ready for full system testing.
- End of Week 11: Testing complete and release-blocking defects resolved.
- End of Week 12: Release package ready for handoff.

The schedule is a starting estimate. Delays in garage data access or third-party integrations may require changes to the planned dates.